

Directorate, Urban Administration & Development, M.P. Bhopal

Format Part-I

Format for Inspection of Ongoing/Completed Work

PART I– Work Information *(To be filled-up by ULB)*

1. GENERAL:

I. Work is (<i>check✓ any one box</i>)	☐	Ongoing	☐	Completed
II. Date of inspection:				
III. Name of Quality Monitor (SQM/SE/EE):				
IV. Division:		District:		ULB/Ward :
V. Name of Road:				
VI. Sanction length:		km, Bituminous Road:		km, Concrete pavement:
VII. Executed length:		km, Flexible pavement:		km, Rigid/Semi-Rigid pavement:
<i>(in case of completed works only)</i>				
VIII. Reasons for deviation (<i>if any</i>):				
IX. Estimated cost (as Sanctioned by UADD):				
	Rs.Lakh			
X. Technical sanction Cost:				
	Rs.Lakh			
XI. Awarded cost:(Financial approval)				
	Rs.Lakh			
XII. Expenditure: (<i>if work is ongoing</i>)				
a. Expenditure done:				
	Rs.Lakh			
b. Bills pending:				
	Rs.Lakh			
Total expenditure (a+b)				
	Rs.Lakh			
XIII. Completion cost: (<i>if work is completed</i>)				
	Rs.Lakh			

XIV. The work is a case of (*check ✓ boxes as applicable*)

A) ☐ New Road Length: m

(i) Carriageway width (m):

B) ☐ Renewal Length: m

(i) Carriageway width (m) :

C) ☐ Strengthening Length: m

(i) Carriageway width (m) :

XV. Date of award of work: (date/ month/ year)

----------------------	----------------------	----------------------

XVI. Date of start of work:(date/ month/ year)

----------------------	----------------------	----------------------

XVII. Stipulated date of completion:(date/ month/ year)

----------------------	----------------------	----------------------

XVIII. Actual date of completion: (date/ month/ year)

----------------------	----------------------	----------------------

(in case of completed work):

2. Construction Programme and Physical Progress: (Ongoing/Completed)

Item of work	Units	Quantity As per DPR	Quantity execute	Completed percentage of item	Fill in start and completion dates			Delay in mont
					Progress status	StartDate	Completion Date	
i) Preparatory Work					Due			
					Actual			
ii) Earth Work& Sub grade					Due			
					Actual			
iii) Sub Base in including shoulders (GSB/CRM)					Due			
					Actual			
vi) WBM/ WMM Base Course					Due			
					Actual			
v) Bituminous Base/Wearing Course					Due			
					Actual			

viii) Bituminous Surface Course					Due			
					Actual			
ix) CC Pavement					Due			
					Actual			
x) Signage etc.					Due			
					Actual			

3. QUALITY CONTROL:

I. Location of field laboratory:

II. Geo-tagged photograph of laboratory uploaded on: (date/ month/ year)

III. Reason for delay in establishment of field laboratory(if so):

IV. Details of contractor, executing the work:

Name of contractor	Mobile number

V. Details of contractor's Engineer available at site: (Responsible for maintaining quality control register part-1

Sl. no.	Name of contractor's Engineer at site	Qualification	Mobile number	Duration of posting at site	
				From	To

VI. Details of Head of ULB supervising the work:

Sl. no.	Head of division (Superintendent Engineer /Executive Engineer)	Duration of posting at site	
		From	To

VII. Details of Assistant Engineer of ULB supervising the work and maintaining quality control register Part-II:

Sl. no.	Name of Assistant Engineer	Mobile number	Duration of posting at site	
			From	To

VIII. Details of the Sub Engineer supervising the work:

Sl. no.	Name of Sub Engineer	Mobile number	Duration of posting at site	
			From	To

IX. List of equipments available in field lab:

4. DETAILS OF MIX DESIGN(s) (if provided in the sanctioned project):

If not provided (check ✓ the box)

	Not applicable
--	----------------

Sl. No	Mix Design	Mix Design Strength	Institute/laboratory where mix design was done	Date
I	Cement Concrete M 30/M40			
II	Dense Bituminous Macadam			
III	Semi Dense Bituminous Concrete			
IV	Bituminous Concrete			

Signature of Engineer (ULB) :.....

Name of Engineer(ULB):.....

Mobile No. :.....

Date:

Directorate, Urban Administration & Development, M.P. Bhopal

Format Part-II

Report of State Quality Monitors (SQM)/ Divisional Engineer (SE/EE) PART II– Observations for Ongoing/Completed Work

(To be filled-up by SQM/SE/EE)

1. GENERAL DETAILS:

I. Date of inspection:

II. Name of Quality Monitor (QM)

III. Division:

District:

ULB/Ward:

IV. Name of Road:

V. RD of inspection: From RD:

m

to RD:

m

VI. Physical status of work:
(*check✓ anyone*)

	Ongoing
--	---------

	Completed
--	-----------

VII. Present status of work: (*check✓ boxes as applicable*)

	Earthwork & Sub grade
--	-----------------------

	GSB
--	-----

	Base course
--	-------------

	Surface course
--	----------------

	Shoulder
--	----------

	Cross drainage work
--	---------------------

	Protection work
--	-----------------

	Drains
--	--------

	Finishing stage
--	-----------------

2. QUALITY ARRANGEMENTS-OBSERVATIONS: (in case of ongoing works only):
(Geo-tagged photograph of the laboratory showing the available equipment's to be uploaded)

- I. Whether field laboratory established: **(check ✓ any one box)** ☐ Yes ☐ Partly ☐ No
- II. Whether location of field laboratory is same as indicated by ULB in format part-I: **(check ✓ any one box)** ☐ Yes ☐ No
- III. Whether necessary equipments as indicated in part-1 are actually available: **(check ✓ any one box)** ☐ Yes ☐ Partly ☐ No
- IV. Whether equipment's have been used: **(check ✓ any one box)** ☐ Yes ☐ Partly ☐ No
- V. If all necessary equipments are not available, whether you have verified them with the list of deficient equipment's provided by ULB in format Part-I: **(check ✓ any one box)** ☐ Yes ☐ No
- VI. Whether contractor's engineer as per Part-I of this format, is available at site: **(check ✓ any one box)** ☐ Yes ☐ No
- VII. If contractor's engineer as per Part-I of this report is not available, whether you are satisfied with alternative arrangement made to maintain QCR: **(check ✓ any one box)** ☐ Yes ☐ Partly ☐ No

3. EARTHWORK & SUB GRADE:

I. Compaction for Embankment and Sub-grade Construction:

Maximum dry density (MDD) kN/m³,

Optimum moisture content (OMC)..... % - As per DPR

Table-2

S.No.	Location (RD)	Degree of Compaction (Measured by QM)			
		Field Moisture Content (%)	Dry Density kN/m ³	% Compaction	Grade (S/U)

Guiding note: If compaction at any location is "U", sub-item grading shall be "U".

Sub-Item Grading :

(check ✓ any one box)

☐	S	☐	U
--------------------------	---	--------------------------	---

(If this sub item is graded U, clear reasons and suggestions for improvement are to be recorded)

(S = Satisfactory, U = Unsatisfactory)

4. GRANULAR SUB-BASE (GSB):

- I. Provision made in the sanctioned DPR: (*check ✓ any one box*) ☐ Yes ☐ No
- II. Item execution status: ☐ Completed ☐ Ongoing ☐ Not yet started
(*check ✓ any one box*)
- III. GSB Grading as per DPR: ☐ Grade-I ☐ Grade-II ☐ Grade-II
(*check ✓ any one box*)
- IV. Maximum Dry density- kN/m³ (*as per DPR*)
- V. Optimum Moisture Content- %

Table-3

Observations- Quality of Material and workmanship

Location (RD)	Conforms to Grading (Y/N)	Material Suitable from plasticity angle (Y/N)	Dry density kN/m ³	% Compaction	Whether compaction is adequate (Y/N)	Thickness as per DPR (in mm)	Measured Thickness (in mm)	Prescribed Thickness provided (Y/N)

- VIII. Whether GSB has been constructed in layers: (*check ✓ any one box*) ☐ Yes ☐ No
(if yes, *check ✓ the number of layers*) ☐ In one layer ☐ In two layer
- IX. Whether compaction has been done as per the provision in DPR: (*Check ✓ any one box*) ☐ Yes ☐ No

Item Grading:-

(*Check ✓ any one box*)

☐ S ☐ U

(if this item is graded as U, clear reasons and suggestions for improvement are to be recorded).

(S = Satisfactory, U = Unsatisfactory)

5. BASE COURSE: 1st Layer:

- a. Provision made in the first layer in sanctioned DPR: (*Check ✓ any one box*) ☐ WMM ☐ WBM ☐ Not Provided
- b. Item execution status: ☐ Completed ☐ Ongoing ☐ Not yet started
(*Check ✓ any one box*)

- c. Actual execution in first layer of the Base course: (**Check ✓ any one box**)

☐	WMM	☐	WBM	☐	Not Provided
--------------------------	-----	--------------------------	-----	--------------------------	--------------

- d. Reason for change in actual execution at site
w.r.t provision made in DPR: (if applicable)

--

- e. Observations- Quality of Material and Workmanship:

Table-4

Location (RD)	Grading of Aggregates (S/U)	Plasticity of Filler material (S/U)	Volume of filler material percent of coarse Aggregate	Compaction based on volumetric analysis (S/U)	Design thickness as per DPR (mm)	Thickness of each layer of WBM/ WMM (mm)	Thickness adequate (S/U)

Item Grading- :

(**Check ✓ any one box**)

(This item is graded as U, clear reasons and suggestions for improvement are to be recorded).

☐	S
--------------------------	---

☐	U
--------------------------	---

(S = Satisfactory, U = Unsatisfactory)

6. BASE COURSE: 2nd Layer:

- a. Provision made in the second layer in the sanctioned DPR: (**check ✓ any one box**)

☐	WMM	☐	WBM	☐	Not yet started
--------------------------	-----	--------------------------	-----	--------------------------	-----------------

- b. Item execution status:
(**Check ✓ any one box**)

☐	Completed	☐	Ongoing	☐	Not yet started
--------------------------	-----------	--------------------------	---------	--------------------------	-----------------

- c. Actual execution in the second layer of the Base course: (**check ✓ any one box**)

☐	WMM	☐	WBM	☐	Not Provided
--------------------------	-----	--------------------------	-----	--------------------------	--------------

- d. Reason for change in actual execution at site
w.r.t provision made in DPR: (if applicable)

--

e. Observations- Quality of Material and Workmanship:

Table-5

Location (RD)	Grading of Aggregates (S/U)	Plasticity of Filler material (S/U)	Volume of filler material percent of coarse Aggregate	Compaction based on volumetric analysis (S/U)	Design thickness as per DPR (mm)	Thickness of each layer of WBM/ WMM (mm)	Thickness adequate (S/U)

Item Grading:-

(Check ✓ any one box)

(This item is graded as U, clear reasons and suggestions for improvement are to be recorded).

☐ S

☐ U

(S = Satisfactory, U = Unsatisfactory)

7. BITUMINOUS BASE COURSE:

(Please do not fill this section if not provided in the DPR)

a. Provision made in the sanctioned DPR: (check ✓ any one box)

☐ BM

☐ DBM

☐ Not provided

b. Item execution status:

(check ✓ any one box)

☐ Completed

☐ Ongoing

☐ Not yet started

c. Actual execution at the site: (check ✓ any one box)

☐ BM

☐ DBM

d. Thickness of layer as per DPR: mm

e. Type and grade of binder used:

f. Brand name of bitumen:

g. Whether the invoices for the whole quantity of

bitumen used at site are available: (check ✓ any one

box)

☐ Yes

☐ No

- a. If the invoice of sufficient quantity not available reason thereof:

--

- b. Bitumen Content as per DPR: %

- c. Whether mix design is done: (in case of BM or DBM) (*Check✓ any one box*) ☐ Yes ☐ No

- d. Marshal stability as per mix design (in case of BM or DBM): KN

- e. Whether inspection of hot mix plant done by SQM /SE: (*Check✓ any one box*) ☐ Yes ☐ No

Date of inspection:.....

Table-6

Location (RD)	Grading of Coarse Aggregates (S/U)	Laying Temperature of the mix as per Norms	Bitumen content %			Marshal stability as per Norms (KN)	Observed thickness of layer (mm)	
			As per Norms	As measured by SQM	S/U		As measured by SQM	S/U

Item Grading-10:

(*Check ✓ any one box*)

(This item is graded as U, clear reasons and suggestions for improvement are to be recorded).

(S = Satisfactory, U = Unsatisfactory)

8. BITUMINOUS SURFACE COURSE:

- a. Provision made in the sanctioned DPR: (*check ✓ any one box*)

☐ OGPC & seal coat	☐ SDBC	☐ Mix Seal Surface
☐ Surface dressing in one layer	☐ Surface dressing in two layer	☐ Bituminous Concrete

- b. Item execution status: (*check ✓ any one box*) ☐ Completed ☐ Ongoing ☐ Not yet started

- c. Type of bituminous surface executed: (*check ✓ any one box*)

☐ OGPC & seal coat	☐ SDBC	☐ Mix Seal Surface
☐ Surface dressing in one layer	☐ Surface dressing in two layer	☐ Bituminous Concrete

- d. Thickness of layer as per DPR: mm
- e. Type and Grade of Binder used:
- f. Brand name of bitumen supplier: (as per record)
- g. Whether the invoices for the whole quantity of bitumen used at site are available: (check ✓ any one box) ☐ Yes ☐ No
- h. If invoice of sufficient quantity not available reason thereof:
- i. Bitumen Content as per DPR: %
- j. Whether mix design done: (check ✓ any one box) (Only in case of SDBC or BC) ☐ Yes ☐ No ☐
- k. Marshal stability as per mix design: KN (Only in case of SDBC or BC)
- l. Any signs of distress on surface: (check ✓ any one) (If yes, check✓ one or more boxes)
- | | |
|---|---|
| ☐ Due to laying at low temperature | ☐ Due to poor workmanship of base course |
| ☐ Due to over rolling | ☐ Due to less/excess bitumen content |

Table-7

Location (RD)	Grading of Coarse Aggregate (S/U)	Laying Temperature of the mix as per Test result	Bitumen content %			Observed thickness of layer		surface unevenness (S/U)
			As per Test I	As measured by SQM	S/U	As measured by SQM	S/U	

Item Grading:

(Check ✓ any one box)

(This item is graded as SRI/U, clear reasons and suggestions for improvement are to be recorded).

(S = Satisfactory, SRI = Satisfactory Required Improvement, U = Unsatisfactory)

9. SHOULDERS:

- a. Item execution status: ☐ Completed ☐ Ongoing ☐ Not yet started
(check ✓ any one box)

- b. Observations- Quality of Shoulders (in case of completed works only):-

Table-8

Location (RD)	Quality of the Material from hand feel test (S/SRI/U)	Degree of Compaction			Camber		
		As per Test result	As measured by SQM (%)	Grading (S//U)	As per DPR (%)	As measured by SQM (%)	Grading (S//U)

Item Grading:

(check ✓ any one box)

(This item is graded as SRI/U, clear reasons and suggestions for improvement are to be recorded).

☐ S ☐ SRI ☐ U

(S = Satisfactory, SRI = Satisfactory Required Improvement U = Unsatisfactory)

10. CEMENT CONCRETE PAVEMENTS:

- I. Type of Cement Concrete Pavement: ☐ Conventional ☐ Cell filled ☐ Panelled concrete
(check ✓ any one box)

- II. Provision made in the sanctioned DPR: (check ✓ any one box)

☐ DLC ☐ Not yet started

- III. Item execution status: ☐ Completed ☐ Ongoing ☐ Not yet started
(Check ✓ any one box)

- IV. Grade of Concrete as per DPR: ☐ M25 ☐ M30 ☐ M40 ☐ Not yet started
(check ✓ any one box)

- V. Item execution status:
(Check ✓ any one box)

- VI. Expansion Joints Sealing ☐ Completed ☐ Ongoing ☐ Not yet started

- VII. CC pavement length proposed as per sanctioned DPR: Proposed length: (m)
Executed length: (m)

Table-9 : Observation should be made for each portion of CC course

Reference RD of CC pavements (m)		RD at which observation made (m)	Quality of Concrete pavement (Visual inspection) (S/U)	28 days Strength of Concrete as per Norms MPa	Quality of Workmanship wearing surface, joints, edges etc. (S/U)	Thickness		
From	To					As per DPR (mm)	As Measured by SQM (mm)	Acceptable (Y/N)

VIII. Quality of cuts and joints is acceptable: *(check✓ any one box)*

☐	Yes	☐	Partly	☐	No
--------------------------	-----	--------------------------	--------	--------------------------	----

IX. Whether the joints have been properly filled with a sealant
(check✓ any one box)

☐	Yes	☐	Partly	☐	No
--------------------------	-----	--------------------------	--------	--------------------------	----

X. Whether surface texture of the pavement is acceptable:
(check✓ any one box)

☐	Yes	☐	Partly	☐	No
--------------------------	-----	--------------------------	--------	--------------------------	----

XI. Whether any honeycombing observed on edges of pavement:
(check✓ any one box)

☐	Yes	☐	Partly	☐	No
--------------------------	-----	--------------------------	--------	--------------------------	----

XII. Whether adequate camber is provided:
(check✓ any one box)

☐	Yes	☐	No
--------------------------	-----	--------------------------	----

XIII. Check strength with rebound hammer; is it acceptable:
(check✓ any one box)

☐	Yes	☐	No
--------------------------	-----	--------------------------	----

Item Grading:-*(Check ✓ any one box)*

☐	S	☐	U
--------------------------	---	--------------------------	---

(If This item is graded as U, clear reasons and suggestions for improvement are to be given)

(S = Satisfactory, U = Unsatisfactory)

11. QUALITY GRADING OF ITEMS AND SUB-ITEMS OF WORK:

1. General details				
--------------------	--	--	--	--

Item 1– Geometrics				
a	Road way width	Ongoing / complete	S/U	
b	Carriageway width		S/U	
c	Camber		S/U	
d	Super elevation	Ongoing / complete	S/U	
e	Extra Widening at Curves	Ongoing / complete	S/U	
f	Longitudinal Gradient in case of road in hilly/ rolling terrain.	Ongoing/ complete	S/U	
	Item Grading		S/U	

Item 2 – Granular Sub-Base (GSB)				
a	Grain Size	Ongoing or complete	S/U	
b	Plasticity		S/U	
c	Compaction	Ongoing or complete	S/U	
d	Total Thickness of Layer	Ongoing or complete	S/U	
	Item Grading		S/U	

Item 3 - Base Course – Water Bound Macadam (WMM) 1st Layer				
a	Grain Size of Course Aggregate	Ongoing or complete	S/U	
b	Plasticity of Crushable Aggregate used as fillers		S/U	
c	Adequacy of Compaction through Volumetric analysis.		S/U	
d	Thickness of every layer of WBM.		S/U	
	Item Grading		S/U	

Item 4 - Base Course – Water Bound Macadam (WMM) 2nd Layer				
a	Grain Size of Course Aggregate	Ongoing or complete	S/U	
b	Plasticity of Crushable Aggregate used as fillers		S/U	
c	Adequacy of Compaction through Volumetric analysis.		S/U	
d	Thickness of every layer of WBM.		S/U	
	Item Grading		S/U	

Item 5- Bituminous Base Course: Bituminous Macadam (BM) and Dense BM				
a	Grading of Coarse Aggregate	Ongoing / complete	S/U	
b	Bitumen Content		S/U	
c	Thickness of Layer		S/U	
	Item Grading		S/U	

Item 6 - Bituminous Surface Course: – OGPC / Seal coat/ Surface Dressing (SD) / SDBC				
a	Gradation of Aggregate	Ongoing /complete	S/U	
b	Laying Temperature of Mix.	Ongoing /complete	S/U	
c	Bitumen content	Ongoing /complete	S/U	
d	Thickness of layer	Ongoing /complete	S/U	
e	Surface Evenness	Ongoing /complete	S/U	
	Item Grading		S/U	

Item 7 – Shoulders				
a	Quality of material for shoulders	Ongoing /complete	S/SRI/U	
b	Degree of compaction	Ongoing /complete	S/SRI/U	
c	Camber.	Ongoing /complete	S/SRI/U	
	Item Grading		S/SRI/U	

Item 8 – Cement Concrete / Semi Rigid Pavements				
a	Quality of Material – Concrete, Coarse Aggregate/ Fine Aggregate.	Ongoing or complete	S/U	
b	Strength of CC in Concrete Pavement		S/U	

c	Quality of Workmanship – Wearing surface texture, Adequacy of setting of concrete, Joints, Sealants, Edges etc.	Ongoing / complete	S/U	
d	Thickness of Layer		S/U	
	Item Grading		S/U	

12. OVERALL GRADING OF WORK:

The overall grading calculated on the basis of item and sub-item wise grading is given below:

#	Item	Awardable Grades	Awarded Grade
1	Quality Control Arrangements	S/ U	
2	Attention to Quality	S/ U	
3	Geometrics	S/U	

4	Granular Sub-base	S/U	
5	Base Course (WMM-II)	S/U	
6	Base Course (WMM -III)	S/U	
7	Bituminous Course (BM and DBM)	S/U	
8	Bituminous Surface Course (OGPC/ Seal coat/ SD/SDBC)	S/U	
9	Shoulders	S/ U	
10	Cement Concrete Pavements	S/U	
Overall Grading		S/ U	

Whether the work can be considered as excellent based on the test results and visual observations made by the quality monitor (*check✓ any one box*)

☐	Yes
--------------------------	-----

☐	No
--------------------------	----

Signature of SQM/Div.(SE/EE):.....

Name:.....

Mobile No. :.....

Date: